We then address the following Research Questions (RQs):

| # | Current top 5 guesses before generating question | | | | | Generated question |
|---|---|---|---|---|---|---|
| 1 | computer | cellphone | book | car | house | G: Is it a living entity? *J: No.* |
| 2 | book | computer | car | building | smartphone | G: Is it man-made? *J: Yes.* |
| 3 | computer | car | building | book | smartphone | G: Can it be held in one hand? *J: Yes.* |
| 4 | smartphone | book | pen | key | mug | G: Is it electronic? *J: No.* |
| 5 | book | pen | key | mug | deck of cards | G: Is it used for writing or drawing? *J: No.* |
| 6 | key | deck of cards | coin | paper clip | dice | G: Is it made of metal? *J: No.* |
| 7 | paper book | wooden spoon | plastic toy | glass cup | porcelain plate | G: Is it used in cooking or eating? *J: No.* |
| 8 | book | playing card | toothbrush | comb | paper clip | G: Is it used for entertainment or leisure? *J: Yes.* |
| 9 | deck of cards | book | puzzle | board game | Rubik's cube | G: Is it associated with a specific sport? *J: No.* |
| 10 | book | deck of cards | dice | puzzle | board game | G: Is it used in board games? *J: No.* |
| 11 | deck of cards | book | paintbrush | puzzle | dice | G: **Is it associated with music?** *J: Yes.* |
| 12 | Harmonica | Guitar pick | Ocarina | Castanets | Recorder | G: Is it a musical instrument? *J: Yes.* |
| 13 | Harmonica | Recorder | Kazoo | Maracas | Tambourine | G: **Is it a stringed instrument?** *J: Yes.* |
| 14 | Ukulele | Violin | Guitar | Mandolin | Harp | G: Is it commonly used in a symphony orchestra? *J: No.* |
| 15 | Guitar | Ukulele | Banjo | Mandolin | Lute | G: Is it a guitar? *J: Bingo!* |

Table 3: Probing the top 5 ranked guesses made by GPT-4 during a game of guessing "Guitar". Some guesses were truncated for readability. **Highlighted turns** are the key turns that eliminates all top 5 guesses.

**RQ1: How does the LLM generate strategies?** We conducted an analysis to probe the internal dialog states underneath the LLMs, to understand how these states evolve throughout the game, and how do they affect the strategy of the model. To this end, at each turn before the model asks a question, we prompt the LLM to predict top 5 candidates given the current dialogue history. This probing process is separate from (thus will not affect) the gameplay. The probing prompt is provided in Appendix C. A GPT-4 game session of guessing "guitar" with dialog state is shown in Table 3. A session on *Celebrities* dataset of guessing "LeBron James" is shown in Table 12 in Appendix C.

From Table 3, it is evident that the model seeks a strategy to ask questions to effectively partition the space among the most promising candidates in the initial 10 turns. After the question is answered, the ranked list of top guesses is adjusted, with certain candidates being eliminated and new candidates being added. For instance, during the third turn, the model asks about the item's feasibility of being held in one hand, enabling the elimination of the larger objects in the leaderboard upon receiving an affirmative answer. during the fourth turn, the model asks if the item is electronic, which ultimately eliminates the top-ranked guess, "smartphone", after receiving a negative response. The remaining items are shifted one position forward to include a new item "deck of cards" to the top-5 guesses.

Interestingly, it is not necessary for the model to always ask questions in order to optimally partition the top-5 candidates. We observed that the strategy also depends on factors including the ease with which a question can be asked to partition the space, and the level of uncertainty the model has about the current top predictions. We provide details on the measurement of uncertainty in Appendix D. In situations where the model is uncertain, it may occasionally *backtrack* and *reexamine* categories that were previously missed. For instance, during the 11th turn, question was asked that ruled out all of the top candidates. This could be due to the fact that the top items are similar in nature, and the model realize there is a significant proportion of other classes that have been overlooked. The successful questioning in these cases led to the recovery of these overlooked classes.

To summarize, a strong G typically exhibits the following behavior: 1) **prioritizing high-level questions** before addressing specific details and enumerations, 2) being aware of the current state and asking questions to effectively **bi-partition the search space**, and 3) being able to **occasionally backtracking** to consider previously overlooked options.

Additionally, we noticed a high level of consistency in GPT-4's gameplay strategy across multiple repetitions (see example in

| | **Things (↑)** | **Celebs (↑)** |
|---|---|---|
| GPT-4 → Vicuna 7B | 0.10±0.02 | 0.14±0.04 |
| GPT-4 → GPT-4 | **0.26±0.02** | **0.40±0.02** |
| Vicuna 7B → Vicuna 7B | 0.07±0.02 | 0.05±0.02 |
| Vicuna 7B → GPT-4 | 0.08±0.01 | 0.06±0.01 |

Table 4: Ablation on planning and reasoning ability. Numbers are the game scores with 5 repetitions. X → Y indicates using the conversation trajectory from X and use the Y for the last turn.

Appendix L), despite some fluctuation in
the order of the questions asked. This suggests that GPT-4 may rely on its own implicit *taxonomy representation* of entities to make decisions, which remains consistent throughout.

**RQ2: Which one is more important in this task, planning or reasoning?** Planning and reasoning abilities affects different stages of game play. Early questions require careful planning to efficiently partition the space, while late game requires more deductive reasoning skills to make an educated guess. We consequently assume that the last turn would only require reasoning ability as no strategic move is needed. With this assumption, we designed the following experiment to investigate the model's planning and reasoning ability in a finer granularity. Given a stronger model GPT-4 and a weaker model Vicuna 7B, and their respective game play trajectories, we only replay the last turn in each trajectory by swapping the guesser model G.

The results are presented in Table 4. Comparing the GPT-4 $\rightarrow$ Vicuna 7B with Vicuna 7B, we observe that stronger planning ability from GPT-4 results in significant improvement. Regarding different reasoners, Vicuna 7B $\rightarrow$ GPT-4 does not show much improvement over Vicuna 7B. This indicates that planning deficiency could result in an unproductive trajectory, poses significant challenges for reasoning during the final step. Moreover, GPT-4 $\rightarrow$ Vicuna 7B shows regression over GPT-4, emphasizing the importance of reasoning in addition to a strong planning capability. Therefore, it is crucial for both planning and reasoning abilities to be strong and work in synergy to achieve optimal performance. See the full table including the results using human trajectory in the Appendix E.

## 5 Enhancing the Open-source Models via Behavior Cloning

We further investigate the potential of open-source models. We used Behavior Cloning (BC) to distill capabilities exhibited by the stronger models into the smaller Vicuna models. We first collected game demonstrations from the GPT-3.5 model over all 600 training examples from *Things* and *Celebrities*. We then fine-tuned the Vicuna models using these demonstrations. The experimental settings are provided in the Appendix I.

**RQ3: Can smaller open-source models benefit from imitating larger close-source models?** We first fine-tuned the Vicuna 7B model using a total of 600 training instances from both datasets. The results are presented in Table 1 as the **V-FT 7B (All)** model. This yielded a more than 70% improvement in both datasets, indicating that weaker models can follow the stronger models' demonstration to use a more effective strategy to steer the conversation on unseen entities or names. We have also observed that BC training can effectively mitigate undesirable behavior such as repetition.

**RQ4: Should models learn from successful demonstration exclusively?** To answer this, we selected all the game runs of the entities or names where the teacher model finally won the game from both datasets. This subset was then used to train the Vicuna 7B model, similar to the *Rejection Sampling* (Touvron et al., 2023b) . The results, referred to as **V-FT 7B (Suc.)** in Table 1, show that imitating from successful experiences outperforms imitating from all experiences (**V-FT 7B (All)**). It gets close to the teacher model (GPT-3.5) on *Things*. However, when it comes to the *Celebrities* dataset, the improvement is more limited.

**RQ5: Does the model size matter?** We further compared the performance improvement achieved through finetuning for both Vicuna 7B and 13B models. The model finetuned on Vicuna 13B, **V-FT 13B (Suc.)**, also achieved much higher score than Vicuna 13B. However, the improvement was less substantial than the improvement on **V-FT 7B (Suc.)**. Again, the improvement over the *Celebrities* dataset is marginal. We hypothesize that guessing the celebrity requires a more case-specific strategies compared to guessing things, thus likely to be more challenging to be superficially imitated.

## 6 Reinforcement Learning from Game-play (RLGP)

We conducted further experiments to investigate whether the performance could be further enhanced through learning solely from the model's own experience. Drawing inspiration from Reinforcement Learning from Human Feedback (RLHF), we employed Proximal Policy Optimization (PPO) (Schulman et al., 2017) to train the model by playing with with the judge, J, a technique we refer to as

RLGP. We made modifications to the trlX repository (Castricato et al., 2023) to facilitate RLGP training. During training, we assigned the reward defined in Eq. (1), to the final turn of each rollout. Additionally, we assigned an intermediate reward to turns that received a "Yes" response from J. We trained the V-FT 7B (Suc.) and V-FT 13B (Suc.) models in Table 1 on 300 *Things* training dateset. Further experimental details can be found in Appendix K. The results in Table 1 present the performance of the RL-trained models, denoted as **V-RLGP**.

**V-RLGP** models exhibit improvement compared to the vanilla Vicuna models when tested on the in-domain dataset *Things*. **V-RLGP 13B** seems to unlock the potential of the Vicuna model, outperforming the **V-FT 13B** model, matching the performance of the runner-up GPT-3.5. **V-RLGP 7B** also show substantial improvement over **V-FT 7B** model. On the out-domain dataset *Celebrities*, interestingly, **V-RLGP** models achieves some improvement.

## 7 Breakdown Analysis: Do the models agree on successful predictions?

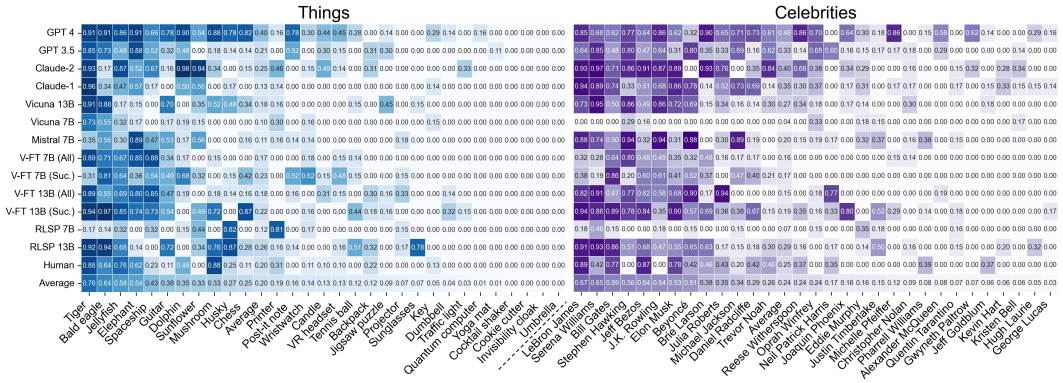


Figure 2: A breakdown of the score of each model on the evaluated items, with the x-axis representing the order of difficulty ranging from easy to difficult. Scores are averaged over 5 repetitions.

We present a comprehensive breakdown analysis of each model's performance on a subset of 30 evaluated items for each dataset in Figure 2. Within the evaluated items of *Things*, four items consistently defy successful predictions by all models. On the other hand, *Celebrities* exhibits a more scattered pattern, with each celebrity being correctly predicted by at least one model.

There are correlations between the entities or names that each model can correctly identify, but different models exhibit their own strengths on different subsets. For instance, the **RLGP 13B** model can accurately predict "Sunglasses", whereas the stronger models like GPT-4 consistently fails. We also provide some case studies including why GPT-4 consistently fails on "Yoga mat" in Appendix L.

We observed that RLGP models tend to strengthen the performance on items that vanilla models occasionally succeed in, thereby improving their success rate on these specific items. However, RLGP models do not effectively facilitate learning about new items. Conversely, BC fine-tuning appears to excel in achieving success on new items. Interestingly, the BC fine-tuned models displays different strengths compared to both the initial checkpoint and the expert whose demonstration it mimics. For example, the **V-FT 13B (Suc.)** model achieves high accuracy in identifying Post-it Note and VR headset, whereas neither the Vicuna 13B nor the GPT-3.5 performs as well in this regard.

## 8 Conclusion

Motivated by a need to develop agents capable of effectively addressing ambiguous user intents, we introduce a testbed for evaluating LLM's strategic planning and deductive reasoning abilities in asking entity-deducing questions. Our findings indicate that SOTA LLMs are able to maintain an intrinsic taxonomic representation of knowledge entities to a certain extent. We further show that this capability can be enhanced through Behavior Cloning or Reinforcement Learning, revealing great potential for further advancements. In future research, we intend to investigate whether the implementation of CoT prompting can further enhance the model's performance in related tasks.